# Cell Biology: Complete Study Notes

This comprehensive guide covers the essential topics in cell biology for undergraduate biology students and health-science trainees. From the fundamental differences between prokaryotic and eukaryotic cells to the molecular machinery of cell signaling, each section is organized with clear explanations, visual aids, and mnemonic devices to help you master the material. Use this document as your primary reference for exam preparation and conceptual understanding.

CELL BIOLOGY | STUDY NOTES | UNDERGRADUATE LEVEL

CHAPTER 1

# Cell Structure & Membrane Biology

All living cells share a common set of features, but they differ dramatically in complexity. **Prokaryotic cells** (bacteria and archaea) lack a membrane-bound nucleus and organelles. Their genetic material exists as a single circular chromosome in a region called the nucleoid. They are typically 1–10 μm in diameter and possess a rigid cell wall (peptidoglycan in bacteria). Ribosomes are smaller (70S) than those found in eukaryotes.

**Eukaryotic cells** (animals, plants, fungi, protists) are defined by membrane-bound organelles, including a true nucleus housing linear chromosomes. They are typically 10–100 μm in diameter, possess 80S ribosomes, and have a complex cytoskeleton enabling intracellular transport and cell shape changes. Plant cells uniquely have chloroplasts, large central vacuoles, and a cellulose cell wall.

### Prokaryotic Cells

- No membrane-bound nucleus
- 70S ribosomes
- Single circular chromosome
- Cell wall: peptidoglycan
- Size: 1–10 μm
- No cytoskeleton

### Eukaryotic Cells

- True membrane-bound nucleus
- 80S ribosomes
- Multiple linear chromosomes
- Membrane-bound organelles
- Size: 10–100 μm
- Complex cytoskeleton

## Plasma Membrane Models

The plasma membrane is a selectively permeable barrier composed of a **phospholipid bilayer** with embedded proteins, cholesterol, and carbohydrates. The **Fluid Mosaic Model** (Singer & Nicolson, 1972) remains the accepted model: phospholipids and proteins move laterally within the plane of the membrane, giving it fluidity. Cholesterol modulates fluidity — it prevents packing at high temperatures and prevents solidification at low temperatures.

Membrane proteins are classified as **integral** (spanning the bilayer, e.g., ion channels, transporters) or **peripheral** (associated with membrane surfaces, e.g., signaling enzymes). The **glycocalyx** — carbohydrate chains on glycoproteins and glycolipids — mediates cell recognition and adhesion. Membrane asymmetry is critical: phosphatidylserine is normally restricted to the inner leaflet; its externalization signals apoptosis.

### Key Membrane Components

- Phospholipid bilayer (amphipathic)
- Cholesterol (fluidity buffer)
- Integral membrane proteins
- Peripheral membrane proteins
- Glycoproteins & glycolipids
- Glycocalyx (cell surface coat)

### Membrane Fluidity Factors

- **Temperature ↑** → fluidity ↑
- **Saturated fatty acids** → rigid membrane
- **Unsaturated fatty acids** → fluid membrane
- **Cholesterol** → stabilizes fluidity
- **Short fatty acid chains** → more fluid

**Mnemonic — FAME: F**luid **A**ssembly of **M**embrane **E**lements (Fluid Mosaic Model)

# Transport Across the Plasma Membrane

Cells must regulate the movement of substances across their membranes to maintain homeostasis. Transport is classified as **passive** (no energy required, down the concentration gradient) or **active** (ATP or electrochemical gradient required, against the gradient). The direction and mechanism of transport depend on the chemical nature of the solute and the electrochemical gradient.

1

### Simple Diffusion

Small, nonpolar molecules ($O_2$, $CO_2$, ethanol) pass directly through the lipid bilayer down their concentration gradient. Rate depends on concentration gradient, lipid solubility, molecular size, and membrane thickness. No protein required.

2

### Osmosis

Passive diffusion of water across a semipermeable membrane from low solute concentration to high solute concentration. Aquaporins (water channels) greatly accelerate osmosis. Critical in kidney function and red blood cell volume regulation.

3

### Facilitated Diffusion

Polar or charged molecules (glucose, amino acids, ions) use membrane proteins to cross down their gradient. **Channel proteins** form aqueous pores (e.g., $K^+$ leak channels, aquaporins). **Carrier proteins** undergo conformational changes (e.g., GLUT1 glucose transporter). Exhibits saturation kinetics (Vmax).

4

### Active Transport

Moves substances *against* their gradient using energy. **Primary active transport** uses ATP directly (e.g., $Na^+/K^+$-ATPase pumps 3 $Na^+$ out, 2 $K^+$ in). **Secondary active transport** uses an existing gradient (e.g., $Na^+$/glucose symporter uses the $Na^+$ gradient to import glucose).

### $Na^+/K^+$-ATPase

Pumps 3 $Na^+$ out / 2 $K^+$ in per ATP. Maintains resting membrane potential. Consumes ~30% of cellular ATP.

### $Ca^{2+}$-ATPase

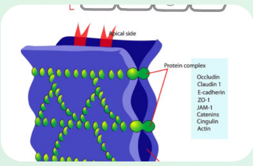
Pumps $Ca^{2+}$ out of cytoplasm into ER or extracellular space. Critical for muscle relaxation and signaling termination.

### $H^+/K^+$-ATPase

Proton pump in gastric parietal cells. Secretes HCl into stomach lumen. Target of proton pump inhibitors (omeprazole).

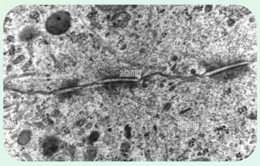
# Cell Junctions: Connecting Cells Together

Cells in tissues are not isolated — they communicate and adhere through specialized junctional complexes. These structures are critical for tissue integrity, barrier function, and intercellular communication. Three major types exist in animal cells: tight junctions, desmosomes, and gap junctions, each with distinct molecular composition and function.



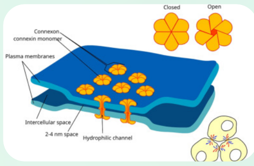

## Tight Junctions (Zonula Occludens)

Tight junctions form a continuous seal between adjacent epithelial cells, preventing paracellular passage of molecules. Key proteins include **claudins** and **occludins**. They also maintain apical-basal polarity by preventing diffusion of membrane proteins between domains. Critical in the blood-brain barrier and intestinal epithelium.



## Desmosomes (Macula Adherens)

Desmosomes are spot welds that anchor adjacent cells together through **cadherins** (desmoglein, desmocollin) linked to **intermediate filaments** (keratin in epithelia, desmin in muscle) via plaque proteins (desmoplakin, plakoglobin). Provide mechanical strength to tissues under stress — skin, heart muscle, bladder.




## Gap Junctions

Gap junctions are channels formed by two **connexons** (each made of 6 connexin proteins) that dock between adjacent cells, creating a pore (~1.5 nm) allowing passage of ions and small molecules (up to 1 kDa). Enable electrical coupling in cardiac muscle and chemical signaling in many tissues. Blocked by low pH and high $Ca^{2+}$.

**Mnemonic — JDG: J**unction types: **D**esmosomes (glue), **G**ap junctions (communication). Tight junctions = seal. Remember: **T**ight = **T**ape seal, **D**esmosome = **D**ouble-sided tape spot, **G**ap = **G**ateway.

CHAPTER 2

# Cell Organelles

Eukaryotic cells compartmentalize functions into membrane-bound organelles, each with a specialized role. This division of labor allows incompatible biochemical reactions to occur simultaneously and increases metabolic efficiency. Understanding organelle structure-function relationships is fundamental to cell biology and pathology.

## Endoplasmic Reticulum

**Rough ER** is studded with ribosomes and synthesizes secretory, membrane, and lysosomal proteins. Proteins enter the ER lumen for folding and glycosylation. **Smooth ER** lacks ribosomes and functions in lipid synthesis, steroid hormone production, detoxification (cytochrome P450), and $Ca^{2+}$ storage (sarcoplasmic reticulum in muscle). The ER is continuous with the outer nuclear envelope.

## Golgi Apparatus

The Golgi is a stack of flattened cisternae (cis, medial, trans faces) that modifies, sorts, and packages proteins from the ER. Key modifications include **glycosylation** (adding/modifying sugar chains), **sulfation**, and **phosphorylation**. The **trans-Golgi network (TGN)** sorts cargo into vesicles destined for lysosomes (mannose-6-phosphate tag), plasma membrane, or secretion. Cis face receives; trans face dispatches.

## Lysosomes

Lysosomes are acidic organelles (pH ~4.5–5.0) containing >60 hydrolytic enzymes (proteases, lipases, nucleases, glycosidases). The acidic pH is maintained by a **V-type $H^+$-ATPase**. They degrade materials via **autophagy** (self-digestion of organelles), **phagocytosis** (engulfed pathogens), and **endocytosis** (receptor-mediated uptake). Lysosomal storage diseases (e.g., Tay-Sachs, Gaucher) result from enzyme deficiencies.

## Mitochondria

Mitochondria are the "powerhouses" of the cell, generating ATP through oxidative phosphorylation. They have a double membrane: the **outer membrane** (porins, permeable to small molecules) and the **inner membrane** (cristae, contains ETC complexes and ATP synthase). The **matrix** contains mitochondrial DNA, ribosomes, and TCA cycle enzymes. Mitochondria replicate independently and are maternally inherited.

## Peroxisomes

Peroxisomes are single-membrane organelles that carry out oxidative reactions using molecular oxygen. They contain **catalase** and **oxidases** that produce and decompose $H_2O_2$. Key functions include **β-oxidation of very-long-chain fatty acids**, detoxification of ethanol (liver), and synthesis of **plasmalogens** (myelin membrane lipids). Defects cause Zellweger syndrome and adrenoleukodystrophy.

# Endosymbiotic Theory

The **Endosymbiotic Theory**, championed by Lynn Margulis (1967), proposes that mitochondria and chloroplasts originated from free-living prokaryotes that were engulfed by an ancestral eukaryotic cell and established a mutually beneficial relationship. Over evolutionary time, these endosymbionts became integrated organelles, transferring many genes to the host nucleus while retaining a reduced genome.

## Evidence Supporting Endosymbiosis

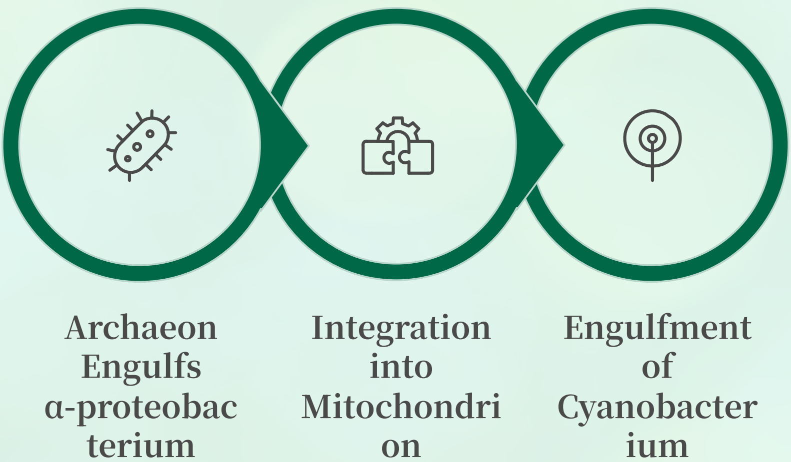
- Mitochondria/chloroplasts have their own **circular DNA** (like bacteria)
- They possess **70S ribosomes** (bacterial-type, not 80S)
- They replicate by **binary fission** independently
- Double membrane structure (inner = original bacterial membrane)
- Antibiotics targeting 70S ribosomes inhibit mitochondrial protein synthesis
- Mitochondrial DNA is **maternally inherited**
- Phylogenetic analysis shows mitochondria are related to **α-proteobacteria**

## Primary vs. Secondary Endosymbiosis

**Primary endosymbiosis** gave rise to mitochondria (from α-proteobacteria) and chloroplasts in plants (from cyanobacteria). These organelles have two membranes.

**Secondary endosymbiosis** occurred when a eukaryote engulfed another eukaryote that already had a chloroplast. This produced organisms with chloroplasts surrounded by **three or four membranes** (e.g., euglenoids, dinoflagellates).

**Gene Transfer:** Most original endosymbiont genes (~90%) migrated to the nuclear genome over time. The organelles now depend on nuclear-encoded proteins imported from the cytoplasm.




The diagram above illustrates the sequential endosymbiotic events: first, an archaeal host cell engulfed an aerobic α-proteobacterium, which evolved into the mitochondrion; later, some eukaryotic lineages engulfed a cyanobacterium, giving rise to chloroplasts in plants and algae.

CHAPTER 3

# Cell Bioenergetics: Respiratory Chain & Chemiosmosis

Cellular respiration extracts energy from nutrients and stores it as ATP. The **electron transport chain (ETC)** and **oxidative phosphorylation** occur at the mitochondrial inner membrane and produce ~34 of the 38 ATP molecules generated per glucose molecule. This process is the primary energy source for aerobic organisms.

## The Respiratory Chain (ETC)

The ETC consists of four protein complexes and two mobile electron carriers embedded in the inner mitochondrial membrane. Electrons from NADH and $FADH_2$ (generated in glycolysis, TCA cycle, and β-oxidation) flow through the complexes, releasing energy used to pump protons across the membrane.

1

### Complex I (NADH Dehydrogenase)

Accepts electrons from NADH → CoQ. Pumps 4 $H^+$. Inhibited by rotenone.

2

### Complex II (Succinate Dehydrogenase)

Accepts electrons from $FADH_2$ → CoQ. Does NOT pump protons. Part of TCA cycle.

3

### Complex III (Cytochrome $bc_1$)

CoQ → cytochrome c. Pumps 4 $H^+$. Q-cycle mechanism. Inhibited by antimycin A.

4

### Complex IV (Cytochrome c Oxidase)

Cytochrome c → $O_2$ (final acceptor, forms $H_2O$). Pumps 2 $H^+$. Inhibited by cyanide, CO.

## Chemiosmotic Hypothesis

Proposed by **Peter Mitchell (1961, Nobel Prize 1978)**, the chemiosmotic hypothesis explains how the ETC drives ATP synthesis. As electrons flow through the ETC, protons are pumped from the matrix to the intermembrane space, creating an **electrochemical gradient** (proton motive force, Δp). This gradient has two components: a chemical gradient (ΔpH) and an electrical gradient (ΔΨ, membrane potential ~–180 mV).

**ATP synthase (Complex V)** uses the proton gradient to synthesize ATP. Protons flow back into the matrix through the $F_0$ subunit (membrane-embedded proton channel), driving rotation of the γ-subunit. This mechanical rotation induces conformational changes in the $F_1$ subunit (catalytic head), driving ATP synthesis via the **binding change mechanism** (Boyer). Approximately 3–4 $H^+$ are required per ATP synthesized.

### ATP Synthase Structure

- **$F_0$ subunit**: membrane-embedded proton channel (c-ring + a subunit)
- **$F_1$ subunit**: catalytic head in matrix ($\alpha_3\beta_3\gamma\delta\varepsilon$)
- **γ-subunit**: rotating stalk connecting $F_0$ and $F_1$
- Each 360° rotation produces ~3 ATP

### Key Inhibitors

- **Oligomycin**: blocks ATP synthase
- **2,4-DNP**: uncoupler (dissipates $H^+$ gradient as heat)
- **Cyanide / CO**: block Complex IV
- **Rotenone**: blocks Complex I
- **Antimycin A**: blocks Complex III

# Cytoskeleton & Nuclear Biology

The cytoskeleton is a dynamic network of protein filaments that provides structural support, enables cell movement, and facilitates intracellular transport. Three major filament systems exist, each with distinct composition, dynamics, and functions. Together they form an integrated scaffold that responds to mechanical and chemical signals.

### Microtubules

**Composition:** α/β-tubulin heterodimers polymerize into hollow tubes (25 nm diameter). Exhibit **dynamic instability** — alternating phases of growth and catastrophe (rapid depolymerization). **Plus ends** grow outward from the centrosome; **minus ends** are anchored at the MTOC.

**Functions:** mitotic spindle formation, vesicle transport (kinesin = plus-end directed, dynein = minus-end directed), cilia/flagella structure (9+2 arrangement), cell shape maintenance.

**Drugs:** Taxol stabilizes; colchicine and vincristine depolymerize microtubules (used in cancer chemotherapy).

### Microfilaments (Actin Filaments)

**Composition:** G-actin (globular) polymerizes into F-actin (filamentous) double helices (7 nm diameter). ATP-bound actin adds to the **plus (barbed) end**; ADP-actin dissociates from the **minus (pointed) end** (treadmilling).

**Functions:** cell motility (lamellipodia, filopodia), cytokinesis (contractile ring), muscle contraction (actin-myosin interaction), microvilli structure, cell cortex rigidity.

**Regulation:** ARP2/3 complex nucleates branched networks; formins nucleate linear bundles; profilin, cofilin, and capping proteins regulate dynamics.

### Intermediate Filaments

**Composition:** Tissue-specific fibrous proteins (10 nm diameter). Most stable cytoskeletal element — no polarity, no motor proteins. Provide tensile strength.

**Types:** Keratins (epithelia), Vimentin (mesenchymal), Desmin (muscle), Neurofilaments (neurons), Lamins (nuclear envelope). Mutations cause diseases (e.g., epidermolysis bullosa from keratin defects).

**Function:** Mechanical integrity, nuclear lamina organization, linking cell-cell junctions (desmosomes) to the cytoskeleton.

## Nuclear Biology

### Nuclear Envelope

Double membrane (inner + outer) continuous with rough ER. The **nuclear lamina** (lamin A, B, C) lines the inner membrane, providing structural support and organizing chromatin. Disassembles during mitosis (phosphorylation by CDK1).

### Nuclear Pore Complex (NPC)

~30 different nucleoporins (Nups) form an octagonal channel (~120 MDa). Allows selective transport: small molecules diffuse freely; large molecules require **nuclear localization signals (NLS)** or **nuclear export signals (NES)** and importin/exportin receptors. Ran-GTP gradient drives directionality.

### Nucleolus

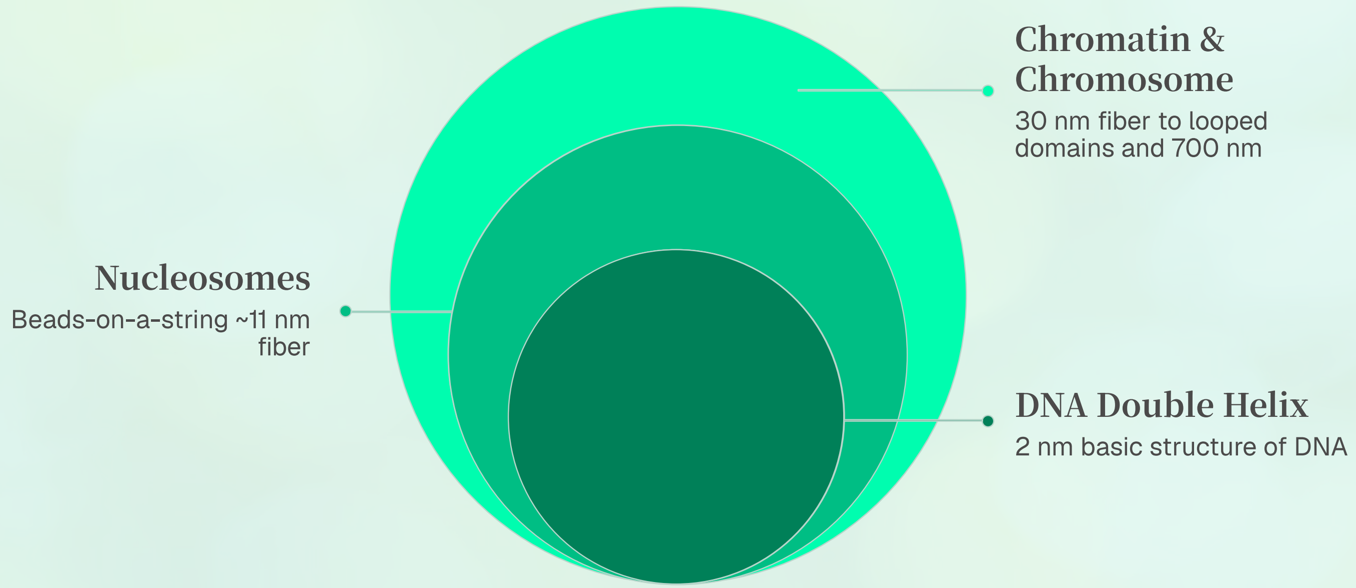
Non-membrane-bound structure within the nucleus. Site of **rRNA transcription** (45S pre-rRNA by RNA Pol I), processing, and ribosomal subunit assembly (40S and 60S). Disassembles during mitosis. Enlarged in rapidly dividing cells and cancer.

### Chromatin States

**Euchromatin:** loosely packed, transcriptionally active, enriched in acetylated histones. **Heterochromatin:** densely packed, transcriptionally silent. **Constitutive** (always silent, e.g., centromeres) vs. **facultative** (conditionally silent, e.g., Barr body/X-inactivation).

## Nucleosome Packaging

DNA is packaged into chromatin through hierarchical folding. The fundamental unit is the **nucleosome**: ~147 bp of DNA wrapped 1.65 times around a histone octamer (two copies each of H2A, H2B, H3, H4). **Linker DNA** (~20–80 bp) connects nucleosomes and binds **histone H1**, promoting higher-order compaction. Nucleosomes fold into a 30-nm fiber (solenoid or zigzag), then into looped domains anchored to a protein scaffold, and finally into metaphase chromosomes.




Histone modifications regulate chromatin accessibility: **acetylation** (by HATs) neutralizes positive charges on lysines, loosening chromatin and promoting transcription; **methylation** can activate or repress depending on the residue and degree (H3K4me3 = active; H3K27me3 = repressed). This "histone code" is read by chromatin remodelers and transcription factors.

CHAPTER 5

# Cell Division: Mitosis, Meiosis & Cell Cycle Regulation

Cell division ensures growth, tissue repair, and reproduction. **Mitosis** produces two genetically identical diploid daughter cells; **meiosis** produces four genetically diverse haploid gametes. Both processes are tightly regulated by cyclin-dependent kinases (CDKs) and checkpoints that ensure fidelity.

### Mitosis ($2n \rightarrow 2n$)

- **Prophase:** Chromosomes condense; spindle forms; nuclear envelope breaks down
- **Prometaphase:** Kinetochores attach to spindle microtubules
- **Metaphase:** Chromosomes align at metaphase plate; checkpoint ensures all kinetochores attached
- **Anaphase:** Sister chromatids separate (cohesin cleaved by separase); move to opposite poles
- **Telophase:** Nuclear envelopes reform; chromosomes decondense
- **Cytokinesis:** Actin-myosin contractile ring divides cell (animals); cell plate forms (plants)

### Meiosis ($2n \rightarrow n$)

- **Meiosis I (reductional):** Homologous chromosomes separate
- **Prophase I:** Synapsis, crossing over (genetic recombination), tetrads form
- **Metaphase I:** Homologous pairs align (independent assortment)
- **Anaphase I:** Homologs separate (sister chromatids stay together)
- **Meiosis II (equational):** Sister chromatids separate (like mitosis)
- **Result:** 4 haploid cells, each genetically unique

**Genetic Diversity Sources:** (1) Crossing over in Prophase I; (2) Independent assortment in Metaphase I; (3) Random fertilization

## Cell Cycle Regulation

The cell cycle is controlled by **cyclin-CDK complexes** that phosphorylate target proteins to drive progression. Different cyclin-CDK pairs act at specific checkpoints:

1

### $G_1$ Checkpoint (Restriction Point)

Cyclin D-CDK4/6 and Cyclin E-CDK2. Checks cell size, nutrients, growth factors, DNA damage. p53 and Rb are key regulators. Most important checkpoint — cells can exit to $G_0$ here.

2

### $G_2$ Checkpoint

Cyclin B-CDK1. Checks DNA replication completion and DNA damage. Prevents entry into mitosis with damaged or unreplicated DNA.

3

### M Checkpoint (Spindle)

Checks all chromosomes are attached to spindle at kinetochores. Prevents aneuploidy. APC/C (Anaphase-Promoting Complex) triggers anaphase onset by degrading securin and cyclin B.

**Tumor suppressors** (p53, Rb, BRCA1/2) halt the cycle upon damage; **proto-oncogenes** (Ras, Myc, Cyclin D) promote progression. Mutations in these genes drive cancer. CDK inhibitors (e.g., p21, p27) provide additional brakes on the cycle.

# Cell Signaling: GPCRs & cAMP Second Messenger

Cells communicate through chemical signals — hormones, neurotransmitters, growth factors, and cytokines — that bind to specific receptors and trigger intracellular responses. Signal transduction pathways amplify and integrate extracellular information to regulate gene expression, metabolism, and cell behavior. Dysregulation of signaling underlies many diseases, including cancer, diabetes, and cardiovascular disorders.

## G-Protein Coupled Receptors (GPCRs)

GPCRs are the largest family of cell surface receptors (~800 in humans) and the target of ~30% of all prescription drugs. They are **seven-transmembrane domain proteins** (7TM) that transmit signals via heterotrimeric G proteins (α, β, γ subunits). In the resting state, Gα binds GDP. Ligand binding causes a conformational change in the receptor, activating the G protein by promoting GDP-GTP exchange on Gα. The GTP-bound Gα and Gβγ dimer dissociate and regulate downstream effectors.

01

### Ligand Binding

Extracellular signal molecule (e.g., epinephrine, glucagon) binds to the GPCR's extracellular domain or transmembrane pocket, inducing a conformational change.

02

### G Protein Activation

Activated GPCR acts as a GEF (guanine nucleotide exchange factor), causing Gα to exchange GDP for GTP. Gα-GTP dissociates from Gβγ dimer.

03

### Effector Activation

**Gαs** stimulates adenylyl cyclase → ↑cAMP. **Gαi** inhibits adenylyl cyclase → ↓cAMP. **Gαq** activates phospholipase C → $IP_3$ + DAG → $Ca^{2+}$ release + PKC activation.

04

### Signal Termination

Gα has intrinsic GTPase activity (accelerated by RGS proteins). GTP hydrolysis returns Gα to GDP-bound state; subunits reassociate. Receptor desensitization via GRKs and arrestins.

## cAMP Second Messenger Pathway

The **cAMP pathway** is a classic GPCR cascade. When Gαs activates **adenylyl cyclase (AC)**, it converts ATP to **cyclic AMP (cAMP)**. cAMP binds to **Protein Kinase A (PKA)**, causing the regulatory subunits to release catalytic subunits. Active PKA phosphorylates target proteins including **CREB** (cAMP Response Element-Binding protein), which translocates to the nucleus and activates transcription of cAMP-responsive genes.

### cAMP Pathway Components

- **Signal:** Epinephrine, glucagon, ACTH
- **Receptor:** GPCR (e.g., β-adrenergic receptor)
- **G protein:** Gαs (stimulatory)
- **Effector:** Adenylyl cyclase
- **Second messenger:** cAMP
- **Kinase:** PKA (Protein Kinase A)
- **Transcription factor:** CREB
- **Termination:** Phosphodiesterase (PDE) degrades cAMP → 5'-AMP

### Clinical Relevance

- **Cholera toxin:** Permanently activates Gαs → uncontrolled cAMP → massive fluid secretion
- **Pertussis toxin:** Locks Gαi in inactive state → unopposed cAMP production
- **Forskolin:** Directly activates adenylyl cyclase (research tool)
- **Sildenafil (Viagra):** Inhibits cGMP phosphodiesterase (PDE5)
- **Caffeine:** Non-specific PDE inhibitor → prolongs cAMP signaling

**Mnemonic — Gs vs. Gi: Gs** = **S**timulates AC (↑cAMP); **Gi** = **I**nhibits AC (↓cAMP). **Gq** = **Q**uestions everything → activates PLC instead.

~800

**Human GPCRs**

Largest receptor family; target of ~30% of all drugs

7

**Transmembrane Domains**

All GPCRs share the 7TM α-helical structure

3

**G Protein Types**

Gαs, Gαi, Gαq — each activates distinct downstream pathways

30%

**Drug Targets**

Approximate proportion of FDA-approved drugs targeting GPCRs